

RAID 0 allows Striping, a scheme where alternate blocks of data is written to different disks. This results in a direct doubling of the write / read speeds. The flip side is that if one of the disks fails, the entire data becomes inaccessible.

RAID 1 offers data redundancy by mirroring the data. In this scheme, the same data is written to both disks. While there is no improvement in performance, from the security standpoint, failure in one disk will not result in downtime, since the same data is mirrored on the other disk(s). Both the above-mentioned schemes require at least two hard disks to function.

RAID 0 + 1 brings together the benefits of both the schemes, and obviously requires a minimum of four hard disks to function. RAID 5 allows striping, but also stores the parity information of every operation, which can be used to recover data in case of failure. This scheme requires a minimum of three drives.

2.1.34 Rotation Speed

This refers to the speed with which the spindle rotates. Naturally, the faster the disk rotates, the faster the data can be accessed and written or read. The fastest hard disks have a speed of 15000 rpm. Regular hard disks are available at 7200 rpm and 5400 rpm, with laptop hard disks also available at 3500 rpm.

2.1.35 SATA

Serial Advanced Technology Attachment refers to the latest form of connection for storage devices like hard disks and optical drives. Data is transferred serially through the interface. The SATA standard has two versions: SATA 1, which offers a maximum data transfer of 1.5 Gigabits per second, and the later SATA 2, which offers data transfer speeds of up to 3 Gigabits per second. Unlike PATA, each SATA channel can connect to only one device.

2.1.1 SCSI

Small Computer System Interface is a high-performance interface mostly used in server systems. It offers better data transfer